



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 17 September 2025, 10:04 AM**State** Finished**Completed on** Wednesday, 17 September 2025, 10:44 AM**Time taken** 40 mins 28 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 int main() {
3     int m;
4     scanf("%d", &m);
5     int arr[m];
6     for (int i = 0; i < m; i++) {
7         scanf("%d", &arr[i]);
8     }
9     int low = 0, high = m - 1;
10    while (low < high) {
11        int mid = (low + high) / 2;
12        if (arr[mid] == 0) {
13            high = mid;
14        } else {
15            low = mid + 1;
16        }
17    }
18    if (low < m && arr[low] == 0) {
19        printf("%d\n", m - low);
20    } else {
21        printf("0\n");
22    }
23    return 0;
24 }
25

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1	0	0	✓

	Input	Expected	Got	
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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